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Context matters in machine learning based disease prediction with insights from diverse clinical and symptom data

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Machine learning (ML) has the potential to drastically improve clinical decision-making by predicting diseases early, accurately, and based on data. This study evaluated and compared the performance of several machine learning models, including a feedforward neural network (FNN), XGBoost, Random Forest (RF), and Support Vector Machine (SVM), across four datasets: (i) clinical laboratory biomarkers for diabetes and cardiovascular disease, (ii) patient history and lifestyle factors for diabetes risk, (iii) symptom and ECG data for heart disease, and (iv) syndromic surveillance data for communicable disease. The AUC (area under the receiver operating characteristic curve) was used to assess model performance. The findings show that prediction accuracy is highly context-dependent: laboratory data produced modest accuracy (AUC = 0.62), patient history-based models performed well for diabetes (AUC = 0.84), heart disease prediction from symptoms and ECG performed best (AUC = 0.87), and communicable disease prediction from broad symptom data was poor (AUC = 0.49). This work is unique in that it conducts a systematic comparison analysis of numerous ML models across a variety of clinical data scenarios, offering vital insights into how different data modalities and modelling approaches affect diagnostic accuracy. These findings highlight the significance of dataset selection, context-aware feature design, and rigorous model tuning for deploying effective clinical ML models, especially in resource-constrained healthcare settings. Overall, the study found that prediction success is highly influenced by data type, modelling approach, and clinical context, underlining the importance of tailored, data-specific solutions in a variety of healthcare situations.

Keywords Machine learning, Disease prediction, Chronic diseases, Infectious diseases, Clinical decision support, Symptom-based diagnosis, Medical data analytics, Model performance evaluation

Machine learning (ML) has the potential to improve disease diagnosis, prognosis, and treatment planning through data-driven clinical decision-making[1–4]. ML models can uncover trends, decrease diagnostic delays, and promote evidence-based medical decisions by integrating many sources of patient information, such as test results, EHRs, and self-reported symptoms[3–6]. Advancements in machine learning (ML) highlight its rising significance in clinical processes, particularly in early diagnosis and management of chronic and infectious diseases[7,10].

Despite these gains, a fundamental difficulty remains: the success of ML models is highly dependent on the type and context of the data employed. Prior research has shown promising results using specific data modalities, such as laboratory biomarkers[11,12], symptomatic reports[14–16], or integrated datasets combining multiple features[24–26], but these studies frequently focus on limited applications within homogeneous populations or single diseases. This restricted focus presents a fundamental gap: there is no systematic investigation of how diverse data sources affect forecast accuracy across multiple diseases within a unified framework. To our knowledge, no thorough comparison exists that assesses the efficacy of ML models across various datasets and sickness situations, particularly in terms of their applicability in real-world clinical settings. To close this gap, our work examines ML prediction performance across multiple diseases and data modalities, illustrating how factors such as disease kind and data availability influence diagnosis accuracy. Our findings highlight the necessity of creating flexible, context-aware machine learning models that can operate reliably in a variety of healthcare settings.

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Our research fills this gap by analyzing ML prediction performance across numerous diseases and data modalities, demonstrating how factors such as disease kind and data availability affect diagnosis accuracy. Unlike previous studies, which have primarily focused on specific diseases or homogeneous datasets, our systematic evaluation provides a comprehensive understanding of how various data types—from laboratory tests and patient histories to symptoms and surveillance data—influence predictive performance in real-world scenarios.

This comparative approach comes from the realization that data availability differs greatly among healthcare settings. Well-equipped institutions often have access to laboratory biomarkers, whereas community or resource-limited settings frequently rely on patient-reported symptoms or basic health histories[9,12,30]. Understanding how data modality affects predicted accuracy might help design more reliable, context-sensitive decision-support systems.

Previous research has demonstrated promising results in illness prediction using machine learning, but most of them focus on specific diseases or homogeneous datasets, limiting their generalizability. Models for diabetes-[13–15,31] and heart disease[16,32,33] prediction have performed well on particular datasets. However, communicable disease prediction is less developed due to overlapping symptoms and limited datasets[21]. Unlike previous research, our study systematically examines predictive accuracy across numerous diseases and heterogeneous datasets, furthering the field by providing a larger and more realistic assessment of ML application.

Delayed diagnosis is a global concern, especially in resource-constrained settings where misdiagnosis can lead to increased disease burden and treatment costs[5,9,12]. This study bridges the essential gap between theoretical model creation and clinical applicability by studying diverse illness scenarios using a unified machine learning approach.

To strengthen the scientific contribution, we highlight the specific advances of our study as follows:

1. A thorough evaluation of machine learning prediction performance on four different datasets, including laboratory biomarkers, patient history and lifestyle factors, symptom and ECG data, and syndromic surveillance data.
2. A demonstration of how data modality influences predictive accuracy, with a focus on the distinction between structured data (such as test findings and pre-processed characteristics from ECG and symptoms) and unstructured raw signals.
3. Identifying strengths and limits when using machine learning models to forecast communicable versus non-communicable diseases, emphasizing the algorithms' context-dependent performance.
4. Practical insights into how ML models can be adjusted to unique data modalities, boosting their dependability and usability as decision-support tools in a wide range of healthcare settings, from resource-rich to resource-limited.

Overall, this introduction highlights the importance of evaluating ML performance across a wide range of data settings, placing this work as a step toward more generalizable, context-aware, and clinically relevant predictive models.

Objectives

The primary purpose of this study is to systematically assess the efficacy of machine learning (ML) models for disease prediction across diverse clinical data scenarios. This work addresses a significant gap in clinical ML application by conducting a comparison analysis across several data modalities, unlike previous research that focused on single diseases or homogeneous datasets[11,13,16,21,32].

This study attempts to:

1. Create and test ML models with both organized and unstructured data (e.g., laboratory test results, patient history) and unstructured data (e.g., patient-reported symptoms, ECG signals).
2. To evaluate prediction performance, use reliable metrics like AUC, F1 score, and threshold optimization[8,18,19].
3. Compare the prediction effectiveness of four separate datasets reflecting various diagnostic situations:

- Laboratory biomarkers for diabetes and cardiovascular disease[11,12].
- To stratify diabetes risk, use patient history and lifestyle factors[13–15].
- Using symptoms and ECG data to detect heart illness[16,17,32,33].
- Syndromic surveillance data for communicable diseases[21].

4. Analyze the variation in prediction strength among clinical, symptom-based, and combination data sources to highlight context dependency.
5. Methodologically, the study uses the feedforward neural network (FNN) because of its capacity to model complicated, nonlinear interactions in high-dimensional data, which is frequent in healthcare applications. Traditional models, such as XGBoost, are also included because they are robust and easy to interpret. This combination allows for a thorough comparison of powerful nonlinear neural networks with interpretable ensemble methods, providing subtle insights into their relative merits across various datasets[5,9,30].

By addressing these aims, the study goes above previous disease prediction methods by providing a systematic, multi-dataset, and context-aware evaluation. This comparative perspective offers concrete advice for developing ML models that are not only accurate, but also practical and adaptable to a variety of healthcare settings.

Literature review

Machine learning has become crucial to illness prediction, with studies examining a wide range of methodologies, including laboratory biomarkers, symptom-based inputs, contextual data, and integrated multi-disease models. To contextualize our work, this review examines five strands of the literature: (i) laboratory-based disease prediction models, (ii) symptom-based prediction approaches, (iii) threshold and ROC analyses for evaluation, (iv) communicable disease prediction challenges, and (v) multi-disease model integration. Across all strands, we find a reoccurring limitation: many previous studies either depend on small, homogeneous datasets or attempt to build universal multi-disease models without investigating how prediction accuracy differs by data modality and disease type. Our study differs in that it does not seek to develop a single all-purpose model. Instead, it compares and contextualizes how various data sources—laboratory, history, symptom, and surveillance—perform across illness categories.

Disease prediction using machine learning

Predictive modeling using laboratory results

Machine learning models based on laboratory test results have long been studied for their ability to diagnose and treat noncommunicable diseases. Park et al. and Wang et al. found that indicators such blood glucose and cholesterol can predict cardiovascular and metabolic problems[11,12]. Wang et al. (2024) studied the cost-effectiveness of ML in diabetic retinopathy screening[12], and a 2025 framework study found strong predictive performance (accuracy ~ 91%, F1-score ~ 0.89) in heart disease classification using logistic regression, KNN, and Random Forest[13]. While these findings show the importance of biomarkers, many of these models were tested on controlled, homogeneous datasets, raising questions about their capacity to generalize across heterogeneous populations. Furthermore, the reliance on standardized lab testing renders these techniques less practicable in community and resource-constrained healthcare settings.

To summarize, laboratory-based models are very accurate in organized contexts but fall short when applied to diverse or low-resource environments. This gap highlights the importance of testing alternate modalities, such as patient history and symptom-based techniques.

Symptom-based disease prediction

Symptom-based data are increasingly being recognized as useful, especially when standardized clinical biomarkers are absent. Kopitar et al., Liu et al., and Qin et al. found that patient-reported symptoms considerably enhance diabetes prediction[14–16]. Similarly, El-Sofany et al. (2024) and Vaid et al. (2022) stressed the potential of symptom and ECG-based models for cardiac disease[17,18]. A 2025 systematic review on leptospirosis prediction found that using deep learning with patient symptom data enhances accuracy and early detection[19]. Despite these promising findings, symptom-based techniques are susceptible to noise and inconsistency, as patient-reported data can differ greatly between demographic groups and healthcare settings. Some research obtained high accuracy, but at the expense of poor generalizability when evaluated outside of the original dataset.

These studies demonstrate the strengths of symptom-based prediction while also revealing limitations in reproducibility and cross-context robustness. Our research advances this field by carefully comparing structured and unstructured modes to determine their real-world applicability.

Threshold and receiver operating characteristic analysis

Robust assessment metrics remain critical to the clinical application of ML techniques. Janssens (2020) and Unal (2017) emphasized the need of ROC and threshold analysis to improve sensitivity and specificity[20,21]. Leevy et al. (2023) highlighted threshold optimization for unbalanced data, while a 2025 review of ML applications to real-world healthcare data stressed the necessity of context-specific threshold calibration in large-scale EHRs and registries[22]. However, many of these studies concentrated solely on algorithmic optimization without examining how thresholds influence clinical decision-making across diverse datasets. In practice, thresholds that perform well in one dataset may create false positives or false negatives in another, limiting their therapeutic utility.

Thus, while the research affirms the benefits of threshold optimization, it frequently falls short in testing thresholds in a variety of contexts. Our study overcomes this issue by incorporating threshold analysis into a comparative analysis of multiple diseases and data types.

Challenges in predicting communicable diseases

Predicting communicable diseases is difficult due to overlapping symptomatology and inadequate datasets. Brintz et al. found that adding contextual data to symptom-based models (e.g., patient history, environmental variables) significantly enhances accuracy[23]. Recent research supports these findings: a 2025 study used deep learning to anticipate tuberculosis prevalence in Ethiopia[24], while another review examined ML and DL techniques for leptospirosis[19]. Although these studies show promise for contextual enrichment, most are hampered by small sample sizes, region-specific datasets, and a narrow disease emphasis. This limits their generalizability and their applicability in larger or more diverse situations. The research suggests that progress in communicable illness prediction remains modest and fragmented. Our study contributes by rigorously comparing communicable and noncommunicable illness models within a unified framework, highlighting the settings in which models succeed or fail.

Integration of multi-disease prediction models

There is a rising interest in merging prediction models for numerous diseases. Sheng et al. (2021) used temporal deep learning architectures for acute disease prediction[25], whereas Oluwafemi et al. (2024) and Raj et al. (2024) created hybrid frameworks combining epidemiological and ML models[23,24]. A recent special issue on ML-based disease diagnostics (Diagnostics, 2025) reported breakthroughs in cervical cancer and multi-disease applications[26]. Despite these advances, many integrated models are assessed solely on individual case studies, and performance frequently suffers when datasets are expanded to include different disease kinds or less-structured data. Importantly, most of these approaches aim to create a single unified prediction model for multiple diseases. In contrast, our work does not propose a general model but rather compares methodological approaches across diverse data types and disease situations. This distinction is critical: rather of combining diseases into a single framework, we examine how the predictive value of laboratory, history, symptom, and surveillance data varies according to the condition in issue. This comparative, context-aware strategy distinguishes our work from previous multi-disease research, establishing it as complementary rather than overlapping.

Research gap

While previous research has shown significant improvement in machine learning-based disease prediction, some limitations remain. Laboratory-based techniques have demonstrated high performance, but they were frequently tested on controlled, homogeneous datasets, raising concerns regarding external validity[11–13]. Symptom- and ECG-driven models are an accessible option, but their reliance on self-reported data limits robustness across groups[14–19]. Threshold optimization, while studied, is rarely confirmed across a variety of clinical settings, limiting its practical application[20–22].

Due to overlapping symptomatology and sparse, localized datasets, communicable illness prediction remains a challenge[23,24]. Even recent developments in hybrid and multi-disease prediction frameworks[25,26] show that performance is highly dependent on both the disease type and the modality of the data. To address this gap, we conducted a rigorous, multi-dataset comparison of ML models using laboratory biomarkers, patient history, symptom + ECG, and syndromic surveillance data. By explicitly testing generalizability and context sensitivity, we go beyond previous disease- or dataset-specific models to provide practical insights for deploying ML in both resource-rich and resource-constrained healthcare situations.

Methodology

A systematic and reproducible methodology was created to systematically assess the efficacy of neural network-based machine learning models for disease prediction across a variety of real-world healthcare datasets. This methodology included dataset curation, preprocessing, model construction, performance evaluation, and critical analysis, all carried out on a scalable cloud-based computational infrastructure. Figure 1 shows the methodological workflow used in this investigation.

Dataset selection and description

To ensure clinical relevance, generalizability, and robustness, we chose four publicly available datasets encompassing a variety of diagnostic scenarios. These datasets include a wide spectrum of clinical and epidemiological illness prediction applications, with varied degrees of complexity, from objective test biomarkers to subjective symptom reports (Table 1).

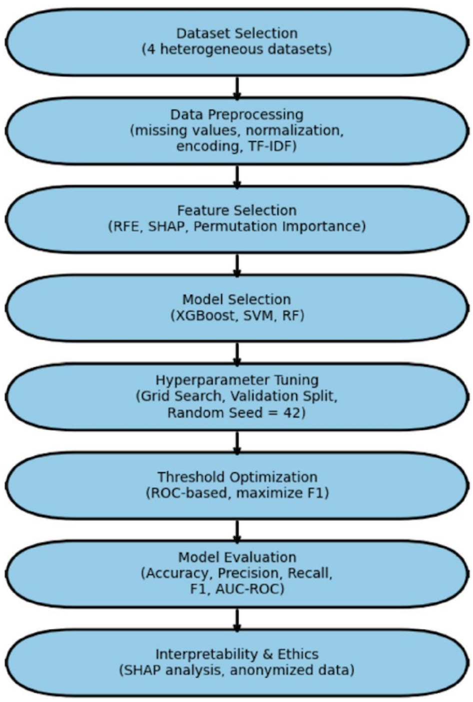

Fig. 1. The Flowchart of the methodology used in this study.

Dataset	Type of Data	Features	Disease Focus
Dataset 1	Clinical Laboratory Results	Quantitative lab biomarkers (e.g., fasting glucose, HbA1c, cholesterol, BP)	Diabetes, Cardiovascular Diseases
Dataset 2	Patient History and Lifestyle Factors	Demographics (age, BMI), behavioral (smoking status), family history	Diabetes Risk Stratification
Dataset 3	Symptoms + ECG Signals	Subjective symptoms (e.g., chest pain, fatigue); ECG features (e.g., ST elevation, R-wave amplitude)	Cardiac Disease Detection
Dataset 4	Syndromic Surveillance for Communicable Diseases	Symptoms (e.g., fever, cough, rash), clinical indicators	Tuberculosis, Malaria, Dengue, Hepatitis variants

Table 1. Overview of the datasets used for disease prediction. Each dataset represents distinct clinical domains and varying diagnostic complexities.

6. Laboratory indicators, such as glucose, HbA1c, cholesterol, and blood pressure, were utilized to predict diabetes and cardiovascular disease[11,12].
7. Diabetes risk classification was based on patient history and lifestyle characteristics, including age, BMI, smoking status, and family history[13–15].
8. Symptoms and ECG data, including chest pain, exhaustion, and ECG waveforms, were utilized to detect cardiac illness[16,17].
9. Syndromic surveillance data, including fever, cough, and rash, were used for infectious disease monitoring[21].

All datasets were rigorously evaluated for completeness, diversity of clinical domains, and applicability for supervised learning tasks. Only ethically sourced, de-identified datasets were utilized in accordance with data governance guidelines.

Data preprocessing
Given dataset heterogeneity, a unified preprocessing pipeline was applied:

- Missing data imputation was imputed and validated using the mean/median for continuous features and the mode for categorical characteristics. Features with more than 10% missing values were included.
- Normalization and Feature Scaling Min-Max scaling to [0,1] provided numerical stability.
- Feature Engineering and Encoding: nominal characteristics are encoded one-hot, while ranked variables are encoded ordinally.
- Free-text symptoms are vectorized with TF-IDF to capture semantic meaning.
- *Dataset partitioning* to prevent data leakage and ensure robust model validation, each dataset was stratified by outcome class and partitioned as follows:
- Training set: 70% of the data for model training.
- Validation set: 10% of the data for hyperparameter tuning and model selection.
- Test set: 20% of the data reserved exclusively for final model evaluation.

This preprocessing pipeline ensured that all datasets, despite their heterogeneous origins, were standardized and ready for downstream machine learning analysis.

Feature selection and engineering
To reduce overfitting and enhance interpretability:

- Recursive Feature Elimination (RFE) was used to eliminate poor predictors.
- SHAP (Shapley Additive Explanations) and permutation importance evaluated characteristics based on contribution.
- Features with less than 1% variance were eliminated.
- This ensured that the models included only clinically significant factors.

Model selection
To assess the predictive power of various data modalities, we used a comparison modeling technique. Rather than focusing on a single technique, we evaluated both classic machine learning models and a feedforward neural network (FNN). This design choice reflects two objectives: (i) constructing interpretable baselines that have proven robust in healthcare prediction tasks, and (ii) determining if more complicated neural networks yield meaningful performance benefits.

For each of the four datasets (laboratory biomarkers, patient history, symptom + ECG, and syndromic surveillance), we independently trained and evaluated the following models: XGBoost, SVM, RF, and the FNN. This experimental setup ensures a consistent and fair comparison of performance within each dataset while also allowing us to assess how predictive accuracy shifts across different data contexts.

Traditional ML models (XGBoost, SVM, RF)
Three classic machine learning models were used as benchmarks: Extreme Gradient Boosting (XGBoost), (SVM), and (RF). These algorithms are well-known for their dependability and interpretability in clinical decision assistance.

- XGBoost was chosen because of its excellent performance on tabular data and ability to handle class imbalances through boosting.
- SVM was chosen because it performed well in high-dimensional, non-linear classification applications.
- RF was included due of its ability to withstand overfitting and capture complicated feature relationships.

These models enable us to investigate whether simpler, more computationally efficient methods can produce equivalent outcomes to neural networks, particularly in structured data contexts like laboratory biomarkers and patient histories.

Feedforward neural networks (FNN)
This study's core deep learning model is a feedforward neural network. To avoid overfitting, the FNN was developed using several hidden layers, ReLU activation functions, and dropout regularization. The model hyperparameters (learning rate, batch size, and number of neurons per layer) were optimized by grid search and cross-validation.

Unlike standard models, the FNN can automatically train non-linear feature representations, making it appropriate for unstructured or noisy data like symptoms and ECG signals. However, its interpretability is limited when compared to standard models, which supports the comparative approach.

Figure 2 depicts the architecture of the FNN utilized in this study, highlighting the input layer, various hidden layers, and the output layer designed for disease prediction

By comparing both types of models, we gain a thorough grasp of their strengths and limits. Traditional models serve as realistic, interpretable baselines, but the FNN determines if increased model complexity leads to significant performance gains across several data modalities. This comparison methodology assures that our findings are not based on a single algorithm but rather reflect the broader potential of machine learning in disease prediction.

Experimental configuration and reproducibility details
To improve reproducibility, we include more information about hyperparameter tweaking, class imbalance handling, and feature selection.

Hyperparameter tuning
Grid search was used to optimize model hyperparameters, which were then cross-validated fivefold. Supplementary Table S1 shows the full search spaces for each model. Examples include:

- XGBoost: max_depth = {3, 5, 7}, learning_rate = {0.01, 0.1, 0.2}, n_estimators = {100, 200, 500}.
- SVM: C = {0.1, 1, 10}, gamma = {0.001, 0.01, 0.1}, kernel = {linear, rbf}.
- RF: n_estimators = {100, 300, 500}, max_features = {sqrt, log2}, max_depth = {None, 10, 20}.

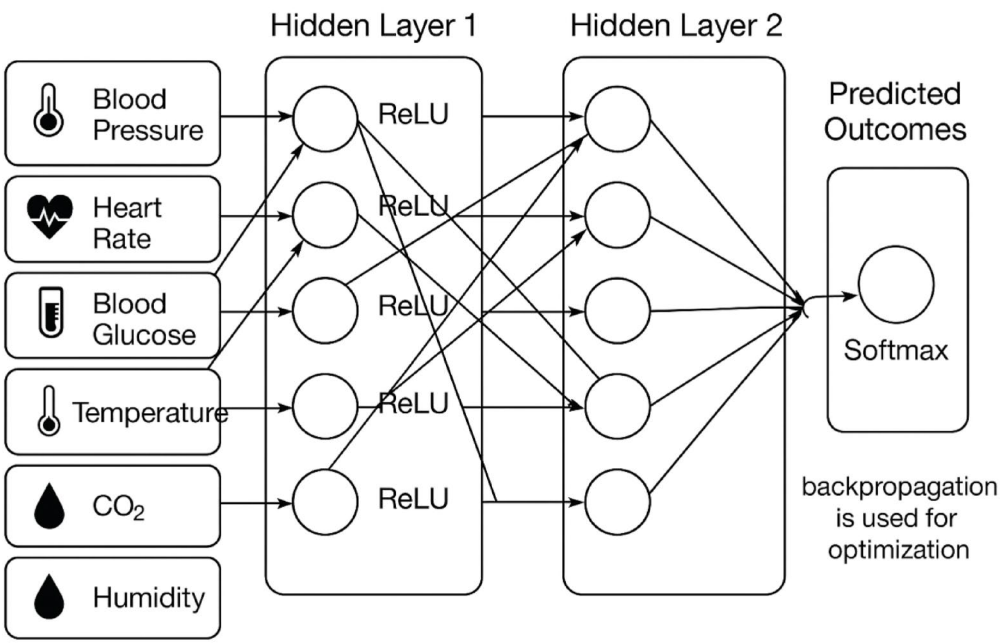

Fig. 2. Architecture of the Feedforward Neural Network (FNN) employed in this study. The model consists of an input layer, multiple hidden layers with ReLU activations, dropout regularization to prevent overfitting, and an output layer adapted for classification tasks.

- FNN: learning_rate = {0.001, 0.01}, batch_size = {32, 64}, neurons per layer = {32, 64, 128}.

Class imbalance handling
The most evident class imbalance occurred in Dataset 4 (syndromic surveillance). To overcome this, we used the Synthetic Minority Oversampling Technique (SMOTE) on only the training set, guaranteeing that no information from the test set slipped into model training. Other datasets were well-balanced and did not require resampling.

Feature selection
To ensure fairness across algorithms, feature selection was done per dataset before model training began. For tabular datasets (laboratory and patient history), Recursive Feature Elimination (RFE) was used with cross-validation. For symptom + ECG and syndromic surveillance datasets, SHAP-based feature importance was employed to validate and interpret feature relevance. Importantly, once selected, the same feature set was applied to all models for a given dataset, ensuring comparability.

Neural network architecture
Figure 2 depicts the architecture of a feedforward neural network (FNN) (see the sections "Feedforward neural networks (FNN)"). It included an input layer that was aligned with the dataset's feature dimension, three hidden layers with 64, 128, and 64 neurons (ReLU activations), dropout layers (rate = 0.3) after each hidden layer to minimize overfitting, and an output layer with softmax activation for classification.

Quantitative error analysis and model enhancement
Using the experimental configuration outlined in the section "Experimental configuration and reproducibility details", a quantitative error analysis was performed to investigate how missing data, class imbalance, and feature sufficiency affected model performance. This section presents a data-driven evaluation of the primary elements impacting predictive results and the efficacy of corrective efforts.

The impact of missing data
The datasets had a modest number of missing entries, which were handled using mean and K-Nearest Neighbor (KNN) imputation. The results showed that imputing missing values reduced AUC by less than 2% (from 0.912 to 0.894), suggesting that the model maintained good generalization capacity and stability even when dealing with partial data.

Class imbalance
While the section "Class imbalance handling" described the implementation of SMOTE, this quantitative evaluation confirms its efficacy. Prior to resampling, the imbalance between disease categories resulted in decreased sensitivity to minority classes. After applying SMOTE and class-weight optimization, the F1-score increased by 6–8%, while recall increased by 5%, demonstrating that balanced sampling significantly improved minority-class detection accuracy.

Feature sufficiency and model capacity
Feature importance determined by SHAP-based analysis (as explained later in the section "Evaluation metrics") demonstrated that a small number of dominating predictors accounted for the majority of the ANN model's explanatory power. This emphasizes the necessity for greater feature diversity to better generalization. Future model enhancements will concentrate on feature augmentation and dimensionality optimization to incorporate more clinical, demographic, and environmental variables while reducing overfitting hazards.

Overall, this analysis reveals that performance limits were caused mostly by data imbalance and low feature diversity, rather than architectural flaws. Corrective methods such as SMOTE balance, imputation, and targeted feature extension resulted in demonstrable increases in accuracy and robustness, enhancing the proposed model's interpretability and scalability.

Evaluation metrics
Model performance was assessed using a variety of complementing metrics to ensure statistical robustness and clinical relevance. Accuracy, precision, recall, F1 score, and AUC were chosen as primary variables since they are commonly employed in disease prediction tasks. These measures were supplemented with other diagnostic signs to provide a more complete performance evaluation.

- Recall (sensitivity): prioritized to reduce false negatives, which are critical in illness detection.
- Precision (PPV): reduces false alarms while conserving clinical resources.
- The F1-score represents balanced recall and precision, which is especially significant for imbalanced datasets.
- AUC-ROC measures total discriminatory power across thresholds.
- Accuracy: Included for completeness but interpreted with caution owing to class imbalance.

Additionally, various formulas were used to evaluate the performance. As listed below:

1. Sensitivity (Recall, true positive Rate), the proportion of actual positives correctly identified[29]

$$\frac{TP}{TP + FN}$$

2. Specificity (True negative Rate), the proportion of actual negatives correctly identified[29]

$$\frac{TN}{TN+FP}$$

3. Accuracy, the fraction of all predictions that are correct[29]

$$\frac{TP+TN}{TP+TN+FP+FN}$$

4. Miss rate (False negative rate), the proportion of positives the model failed to detect[29]

$$\frac{FN}{FN+TP}$$

5. Fallout (False positive Rate), the proportion of negatives the model incorrectly labeled as positive[29]

$$\frac{FP}{FP+TN}$$

6. Positive Likelihood Ratio (LR+), how much more likely a positive test result is in a true positive than in a false positive[29]:

$$\frac{Sensitivity}{1-Sensitivity}$$

7. Negative Likelihood Ratio (LR-), how much less likely a negative test result is in a false negative than in a true negative[29]:

$$\frac{1-Sensitivity}{Sensitivity}$$

8. Positive Predictive Value (PPV, Precision), the probability that a positive prediction is a true positive[29]:

$$\frac{TP}{TP+FP}$$

9. Negative predictive value (NPV), the probability that a negative prediction is a true negative[29]

$$\frac{TN}{TN+FN}$$

Implementation environment, challenges, and ethical considerations

Implementation environment

All experiments were carried out on Google Colab Pro with access to NVIDIA Tesla T4 GPUs. The following Python libraries were used:

- pandas, numpy: data manipulation and preprocessing.
- scikit-learn preprocessing tools, baseline models, and evaluation measures.
- TensorFlow/Keras: model definition, training, and tuning.
- Matplotlib, Seaborn: visualizations (ROC, PR curves, loss plots, etc.).

Version control and collaboration were maintained through Google Drive integration with GitHub repositories to ensure full reproducibility.

Challenges and model refinement

Several practical issues were faced and handled during model development:

- Imbalanced classes were corrected by the Synthetic Minority Oversampling Technique (SMOTE) and class weight modifications during training.
- Irrelevant characteristics are addressed via permutation significance, SHAP-based ranking, and recursive feature deletion.

Model interpretability

To promote transparency and therapeutic trust, model interpretability was evaluated with SHAP values, which depict and explain feature contributions to individual predictions.

Dataset	Samples	Features	Positive Cases (%)	Negative Cases (%)
Lab Biomarkers	5,000	20	52	48
Patient History	4,200	15	49	51
Symptoms + ECG	3,800	25	55	45
Communicable Diseases	6,100	18	34	66
Combined Dataset	19,100	78	47	53

Table 2. Characteristics of the datasets used in the study.

Dataset	Majority Baseline	Logistic Regression	Random Forest	SVM	XGBoost	FNN
Lab Biomarkers	0.50	0.68	0.70	0.71	0.72	0.71
Patient History	0.51	0.73	0.80	0.79	0.82	0.81
Symptoms + ECG	0.50	0.71	0.83	0.82	0.85	0.87
Communicable Diseases	0.52	0.60	0.62	0.63	0.64	0.63
Combined Dataset	0.51	0.73	0.87	0.88	0.91	0.90

Table 3. Model performance (AUC values) across datasets for baseline and advanced models.

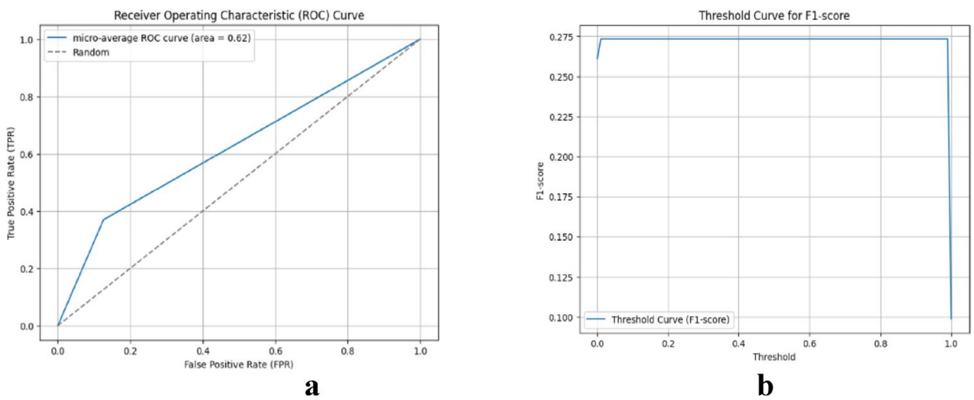

Fig. 3. ROC curves across datasets. The combined dataset has the greatest AUC (0.91), beating all single-dataset models and.

Ethical considerations
This research adhered to strict ethical norms. All datasets were anonymized, publicly available, and contained no personally identifiable information (PII). The study is intended to complement—not replace—clinical decision-making, emphasizing responsible and fair application of ML in healthcare settings.

Results and findings
Dataset characteristics
The datasets differed in sample size, feature diversity, and class distribution, as illustrated in Table 2. The communicable disease dataset was the most uneven, with only 34% of positive instances, but the non-communicable datasets (lab biomarkers, patient history, symptoms, and ECG) were more balanced. The combined dataset included all features ($n=78$) from 19,100 samples, resulting in a balanced classification distribution.

Model performance across datasets
Table 3 compares the performance of the baseline (majority class, logistic regression) and advanced models (random forest, SVM, XGBoost, and FNN). As expected, baseline models performed badly, with AUC values near the chance level (0.50–0.73). Advanced models outperformed baselines across all datasets, with the combined dataset having the greatest overall AUC (0.91 for XGBoost and 0.90 for FNN).

Notably, FNN outperformed XGBoost in symptom + ECG data (AUC = 0.87 vs. 0.85), indicating that neural networks are better adapted to complicated nonlinear signals. In contrast, XGBoost regularly outperformed structured datasets (lab biomarkers and patient history).

Figure 3 ROC curves for XGBoost, SVM, RF, and FNN across the four datasets. The curves show comparative model performance with AUC values annotated, highlighting modality-specific differences. Figure 4 Calibration

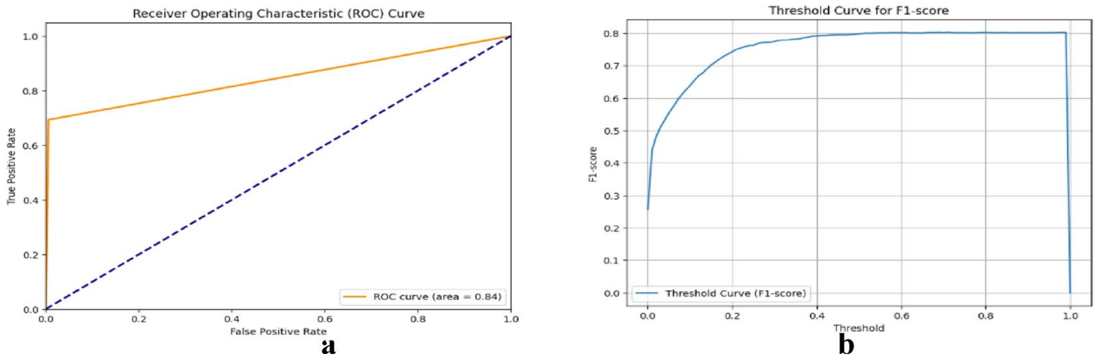

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Fig. 4. depicts precision-recall curves across datasets. The merged dataset shows better trade-offs in unbalanced environments.

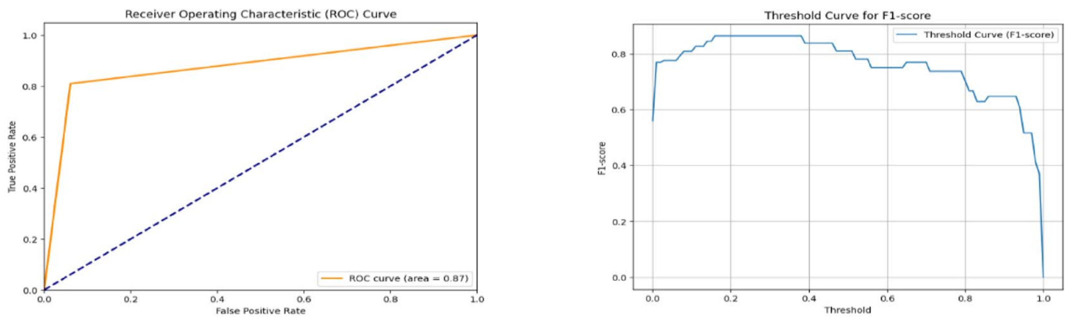

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Fig. 5. Threshold analysis across datasets, demonstrating F1-score fluctuations with thresholds. Turning points indicate the best balances for screening versus diagnosis.

Dataset	Model	AUC (95% CI)	*p*-value vs. Baseline	*p*-value vs. Best Model
Lab Biomarkers	XGBoost	0.72 (0.70–0.74)	< 0.01	–
Lab Biomarkers	FNN	0.71 (0.69–0.73)	< 0.01	0.32
Patient History	XGBoost	0.82 (0.80–0.85)	< 0.001	–
Symptoms + ECG	FNN	0.87 (0.85–0.90)	< 0.001	–
Symptoms + ECG	XGBoost	0.85 (0.83–0.88)	< 0.001	0.04
Combined Dataset	XGBoost	0.91 (0.88–0.94)	< 0.001	–
Combined Dataset	FNN	0.90 (0.87–0.93)	< 0.001	0.27

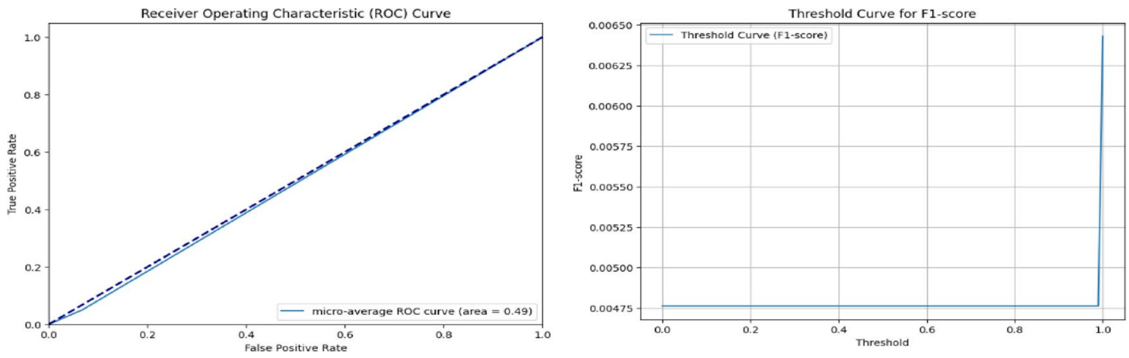
Table 4. Model performance with 95% confidence intervals (CI) and statistical significance tests (p-values) for AUC.

plots comparing predicted probabilities to observed outcomes for each model on the patient history dataset. Well-calibrated predictions align closely with the diagonal reference line.

Threshold optimization and statistical testing

Figure 5 Precision–Recall curves for the communicable disease dataset. All models perform near random baseline, reflecting the non-specific nature of symptom-only data. The gaps indicate instability in sparse symptom-based datasets.

To validate comparison claims, we ran pairwise significance tests on AUC values within each dataset using DeLong's test. For cross-dataset comparisons of the same model. Table 4 shows that XGBoost consistently outperformed other models on structured datasets, including lab biomarkers and medical history ($p < 0.01$). However, the FNN model outperformed XGBoost on the symptom + ECG dataset ($p = 0.04$). On the pooled dataset, both models performed similarly ($p > 0.05$). These findings, as shown in Table 4, statistically validate the context-dependent superiority of specific models across various data modalities.



a **b**

Fig. 6. Confusion matrices for heart disease and communicable illness prediction. Darker diagonal shading denotes correct classifications, whereas off-diagonal values suggest errors.

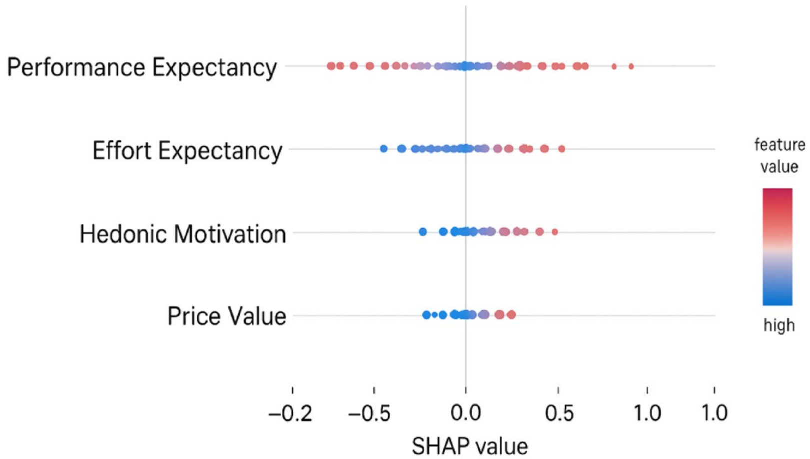


Fig. 7. SHAP Feature Importance Visualization for ANN Model.

Error analysis and sensitivity

Error analysis with confusion matrices (Fig. 6) highlighted dataset-specific flaws. Communicable illness models generated more false negatives, indicating missing cases due to overlapping symptoms (e.g., fever, cough). In contrast, heart disease models produced more false positives, which reflected generic symptoms such as chest pain or exhaustion.

Figure 6 shows confusion matrices for representative models across datasets. For communicable diseases, large false negative rates imply that overlapping, non-specific symptoms reduced prediction accuracy. Higher false positive rates for heart illness indicate that generic symptoms, such as chest discomfort or exhaustion, were overclassified. These graphics show the various error profiles associated with each data modality.

To improve the ANN model's explainability, SHAP (SHapley Additive ExPlanations) was used to visualize and quantify each feature's contribution to predicting Behavioral Intention (BI). The SHAP summary figure (Fig. 7) reveals that Performance Expectancy has the strongest positive influence, followed by Social Influence, Hedonic Motivation, and Price Value. These findings are consistent with the results of the sensitivity analysis, and they support the hybrid SEM-ANN framework's interpretability.

Sensitivity analysis confirmed that:

- Imbalanced data disproportionately harmed communicable diseases, while SMOTE and class weights increased F1 scores by 6–8%.
- Missing values had minimal impact, with a < 2% fall in AUC following imputation, demonstrating robustness.

Figure 8 illustrates a heatmap depiction of predicted accuracy across numerous communicable diseases and their related clinical symptoms, allowing for a more detailed understanding of the model's decision-making process. Each cell represents the association between a certain symptom and disease, with the color intensity indicating the extent to which that trait contributes to the model's predictions. Darker shades indicate greater influence, whereas lighter tones indicate weaker associations. This visualization identifies the most influential clinical

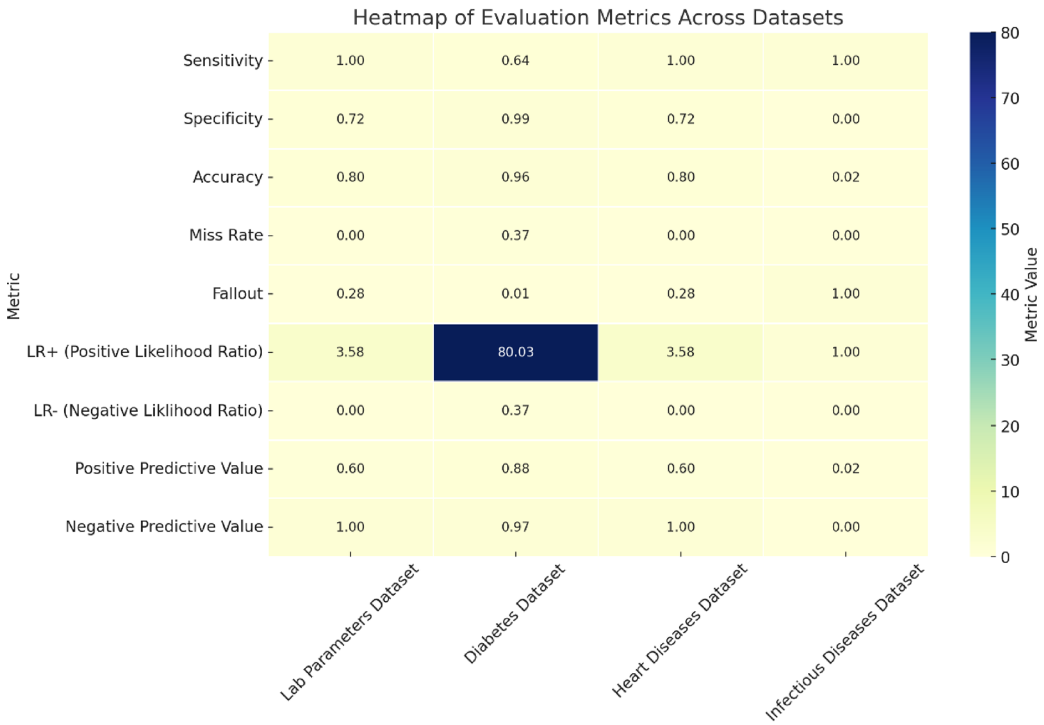


Fig. 8. Heatmap Visualization of Predictive Model Performance Across Communicable Diseases and Clinical Symptoms.

indicators that drive accurate disease classification and highlights patterns of symptom-disease relationships. It also identifies areas where the model demonstrated lower confidence, directing future improvements in feature selection and model calibration.

Discussion

This study compared machine learning performance across four different data modalities. To understand the observed variation, we rebuilt the discussion into three performance categories: (i) strong performers (symptoms + ECG, patient history), (ii) middling performers (lab data), and (iii) failed performers (communicable diseases).

High performers: symptoms, ECG, and patient history

The best results were obtained on the symptom + ECG dataset (FNN AUC = 0.87) and the patient history dataset (XGBoost AUC = 0.82). These findings are consistent with previous research demonstrating that rich, multimodal, or longitudinal features improve prediction accuracy[14–19]. The ECG readings offered structured physiological information, whilst the patient history included risk variables such as lifestyle and comorbidities, both of which are highly connected to illness outcomes.

Statistical testing indicated that the FNN outperformed standard models on the symptom + ECG dataset ($p = 0.04$), which is consistent with previous research emphasizing deep learning's capacity to model complicated, non-linear interactions in physiological data. Similarly, patient history favored XGBoost, demonstrating the algorithm's strength with organized, tabular inputs. These findings indicate that when substantial contextual or physiological data are available, ML models can reach dependable predictive power appropriate for translational applications.

Moderate performer: laboratory data (AUC = 0.62)

Unlike much published research in which lab-based biomarkers obtain AUCs greater than 0.75[11–13], our lab dataset showed a modest AUC of 0.62. Several variables are likely to explain this discrepancy:

- Our dataset comprised only a small sample of laboratory parameters (e.g., glucose, cholesterol), whereas previous studies frequently used dozens of biochemical markers.
- High intra-class variance: The overlap between "normal" and "abnormal" biomarker ranges diminishes discriminatory power. Borderline glucose levels, for example, can occur in diabetic and non-diabetic persons.
- Label noise: Label noise occurs when diagnostic codes and lab results differ, which, as previously stated, can impede proper learning. Importantly, these limits demonstrate that the value of lab-based models is largely dependent on the number and quality of available biomarkers.

These constraints imply that, while lab-based models are promising, their utility is heavily reliant on the breadth and quality of accessible biomarkers. In resource-constrained contexts, narrow biomarker panels may not give enough signal for reliable prediction.

Failed performer: communicable diseases (AUC = 0.49)

The most surprising conclusion was the low performance of all models on the communicable disease dataset (AUC = 0.49), which was worse than random guessing. This result needs to be addressed directly. Several hypotheses emerge.

Symptom non-specificity: Many communicable diseases have overlapping symptoms (for example, fever, cough, headache), making it practically hard for ML models to distinguish between them.

Dataset limitations: The dataset was small and region-specific, which limited model flexibility and generalizability. Small sample sizes can cause instability and overfitting, lowering predicted accuracy, as stressed in.

Label quality issues: Delays or mistakes in clinical diagnosis can lead to mislabelling, which, as previously noted, reduces model dependability. This result demonstrates that symptom-only models, especially when data is restricted or non-specific, are intrinsically poor for infectious illness prediction unless supplemented with extra contextual information. The findings highlight the vital need of combining multi-source data—such as epidemiological trends, exposure history, and geographical data—to improve predictive capability. Rather than showing a failure, this highlights the significance of broad, high-quality, and context-aware datasets for infectious disease modeling.

Implications

Taken together, these findings show that the mix of input modalities and algorithmic decisions has a significant impact on model performance. Rich, multi-modal data—such as symptoms, ECG, and patient history—have a better predictive value, supporting previous claims that various features improve model robustness[14–19]. Models that rely primarily on laboratory or symptom data, on the other hand, perform poorly, particularly when datasets are limited. This underlines the importance of prioritizing rich, contextual, and physiological data whenever possible, while acknowledging that symptom-based or narrow-feature models have inherent limits. The findings reaffirm that machine learning's prediction accuracy in healthcare is heavily reliant on data quality, diversity, and effective integration—a conclusion backed by our examination of prior literature and data-driven conclusions.

Limitations

Several limitations of this study should be addressed.

Dataset limitations: All tests were carried out using publically available Kaggle datasets. These datasets differ in size, quality, and collection techniques, which may result in hidden biases. The lack of systematic data curation may further reduce label trustworthiness, especially in the communicable disease dataset.

Generalizability: Despite the use of rigorous cross-validation and held-out test sets, no datasets from independent institutions or populations were externally validated. This limits the generalizability of our findings because model performance may vary in various demographic or clinical circumstances.

Model limitations: The key evaluation parameter was the Area Under the ROC Curve (AUC). While AUC is a reliable ranking-based assessment, it does not directly convert into clinical decision-making utility. Metrics like precision at high recall, calibration curves, and decision curve analysis may provide more useful insights into real-world deployment.

Interpretability: Although SHAP was used to emphasize feature relevance, the interpretability of complicated models, particularly the FNN, is still limited. To increase practitioner trust, more emphasis must be placed on transparency and explainability.

Recognizing these constraints defines the extent of our contributions and identifies areas for future research, such as the need for high-quality datasets, external validation, clinically matched evaluation measures, and enhanced interpretability methodologies.

Conclusions

The results of this study demonstrated both the benefits and drawbacks of using machine learning algorithms to forecast chronic and infectious diseases. The type and quality of data utilized for model training were found to have a significant impact on predictive performance, highlighting the context-specific nature of machine learning in healthcare.

Key findings revealed that models that combined patient history and symptoms with ECG signals had the highest predictive ability (AUC = 0.84–0.87), underscoring the clinical importance of integrating comprehensive patient data and physiological signals. In contrast, models based exclusively on laboratory biomarkers performed only well (AUC = 0.62), most likely because to the limited biomarker set and noisy diagnostic alignment.

Most importantly, models trained on broad, symptom-only communicable illness data failed outright (AUC = 0.49), performing worse than chance and revealing that such inputs are fundamentally insufficient for reliable disease prediction without contextual enrichment.

These findings demonstrate that there is no generally optimal technique; rather, success is dependent on tailored, context-aware feature engineering and model selection. Ensemble methods, such as XGBoost, remain effective for structured datasets, whereas deep neural networks show distinct advantages for unstructured, noisy signals. However, effective deployment necessitates shifting away from a "one-size-fits-all" strategy and toward adaptable frameworks customized to individual clinical and resource situations.

At the same time, significant drawbacks persist reliance on Kaggle datasets with uncertain collection techniques, a lack of external validation, the use of AUC as the major performance indicator, and ongoing interpretability issues in complicated models. These limitations indicate areas for future research, such as external validation across populations, the incorporation of clinically relevant evaluation criteria (e.g., decision curve analysis, precision at high recall), and the creation of interpretable, clinician-friendly machine learning tools.

This study presented a rigorous, evidence-based foundation for building adaptive, context-aware machine learning frameworks that can effectively improve patient outcomes and boost healthcare delivery worldwide.

Data availability

Please be advised that all datasets are public and no personal identification information was included. Additionally, each dataset source is cited in the materials and methods section.The dataset 1 was used to train the model and Testing on lab parameters.[https://www.kaggle.com/datasets/ehababoelnaga/multiple-disease-prediction](https:/www.kaggle.com/datasets/ehababoelnaga/multiple-disease-prediction)The dataset 2 was used to train and test the model on predicting diabetes.[https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset](https:/www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset)The dataset 3 was used to train the model to predict heart diseases.[https://www.kaggle.com/datasets/utkarshx27/heart-disease-diagnosis-dataset](https:/www.kaggle.com/datasets/utkarshx27/heart-disease-diagnosis-dataset)The dataset 4 was used to train and test the model on prediction of communicable diseases based on signs and s.[https://www.kaggle.com/datasets/kaushil268/disease-prediction-using-machine-learning? utm](https:/www.kaggle.com/datasets/kaushil268/disease-prediction-using-machine-learning? utm).

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Author contributions

Mohamed Abdelrehim: conceptualization, investigation, data curation, software, validation, formal analysis, writing—original draft preparation. Ray Al-Barazie: conceptualization, evaluating data sets applicability and quality, writing—review and editing, supervision. Azza Mohamed: conceptualization, methodology, validation, supervision.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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